Laboration 2: Cellular Automata

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1. Problem statement

In this lab we will be investigating classical examples of cellular automata. Cellular automata are ...

The goal of the lab is to ...

All of these tasks will give us an overview of the characteristics in cellular automata ...

2. Game of Life

2.1. Game of Life as a discrete dynamical system

Conway's Game of Life can be viewed as a discrete dynamical system on a rectangular lattice $X^{(t)} \in \{0, 1\}^{m \times n}$, where each entry $X_{ij}^{(t)}$ represents the state of the cell at site (i, j) at time t. The evolution of the system is determined by update rules that specify the state of each cell at the next time step, $X_{ij}^{(t+1)}$.

Definition 2.1 – Conway's rules [2]

1. **Survival**: a live cell stays alive if it has 2 or 3 live neighbors.
2. **Birth**: a dead cell becomes alive if it has exactly 3 live neighbors.
3. **Death by isolation**: a live cell dies if it has fewer than 2 live neighbors.
4. **Death by overcrowding**: a live cell dies if it has more than 3 live neighbors.
5. **Neighborhood**: the neighbors of a cell are the 8 surrounding cells.

Let $N_{ij}^{(t)}$ denote the degree of node $X_{ij}^{(t)}$ at time t. Since each cell has eight neighbors, the relevant directions are the cardinal and intercardinal directions. Self-interaction is excluded, so the evolution of a cell depends only on its neighborhood. The quantity $N_{ij}^{(t)}$ can be computed as

$$N_{ij}^{(t)} = \sum_{p=-1}^{1} \sum_{q=-1}^{1} K_{p+2,q+2} X_{i+p,j+q}^{(t)}, \tag{2.1}$$

where K is a 3×3 kernel with ones except the center entry that is equal to zero. Equivalently,

$$N^{(t)} = K * X^{(t)}, \tag{2.2}$$

where $K * X^{(t)}$ denotes the convolution of the lattice $X^{(t)}$ with the kernel K.

The update rule can then be written as

$$X_{ij}^{(t+1)} = \begin{cases} 1, & \text{if } X_{ij}^{(t)} = 1 \text{ and } N_{ij}^{(t)} \in \{2, 3\}, \\ 1, & \text{if } X_{ij}^{(t)} = 0 \text{ and } N_{ij}^{(t)} = 3, \\ 0, & \text{otherwise.} \end{cases} \tag{2.3}$$

This shows that the update rule is a non-linear composition of a linear map:

$$X^{(t+1)} = \mathbb{1}\left(N^{(t)} = 3\right) + X^{(t)} \odot \mathbb{1}\left(N^{(t)} = 2\right), \tag{2.4}$$

We use dirichlet boundary conditons or zero padding to handle the adjacency of boundary nodes.

2.1.1. Logic gates in the Game of Life

In classical computation, the basic logic gates `AND`, `OR`, and `NOT` act on one or two binary input streams and produce a binary output stream. They implement a local transformation of information.

In the Game of Life, by contrast, computation is realized not by abstract Boolean operators but by spatially embedded dynamical processes on a lattice. The initial condition is a geometric configuration, and Conway's update rule evolves this configuration through time. The problem of building a logic gate is therefore a problem of geometric design and synchronization: one seeks an initial configuration $X^{(0)}$ that evolves, under the Game of Life rules, into a deterministic spatiotemporal process implementing `AND`, `OR`, or `NOT`.

This is achieved by using gliders as moving signals and glider guns as periodic sources of such signals [3, 4]. Glider trajectories function as wires, constraining the direction of information flow; in the absence of interference, the signal propagates persistently along these paths. To represent binary information, the presence of a glider stream is interpreted as `1`, while its absence is interpreted as `0`. Eaters are used as stabilizing and filtering structures, suppressing selected streams and enabling controlled interactions between multiple inputs. By translating and positioning guns, eaters, and collision regions appropriately, one obtains deterministic and composable gate constructions [1], so that the output of one gate may be routed as the input of another without loss of logical meaning. computational fidelity.

2.1.2. Encoding of Inputs

Definition 2.2 – Encoding binary signals

Signal generation is achieved using the Gosper glider gun, which produces a periodic sequence of gliders with fixed phase and spacing. Through geometric transformations (rotations and reflections), identical guns can be oriented to produce signals in any required intercardinal direction. Each gun occupies a finite region of the lattice (for the canonical gun, 36×9 cells), but its function is not the static occupancy of this region; rather, it continuously emits a sequence of mobile patterns that carry information across the lattice. Consequently, the encoding is fundamentally dynamic and cannot be reduced to a fixed memory footprint.

1. **Signal 1** (`TRUE`): This is represented by the **prescence** of a **glider**. This signal propagates dynamically across a spatial lane across the lattice in fixed intercardinal directions.
2. **Signal 0** (`FALSE`): Defined as the **inexistence** of a **glider**. Its realization depends on the observation point. There are two basic cases:
   - **Blocking**: Physically *obstructing* a Gosper gun.
   - **Annihilation**: Using collision trajectories or *eaters* to remove a glider from the stream

Definition 2.3 – Signal Interaction Mechanics

Logic is performed through the interaction of these signals. Because gliders are sensitive to **synchronization**, the timing of these interactions is as important as their location.

1. **Glider-Glider Collision**: These collisions are different depending on their spatiotemporal initial configuration.
   - Requires: Antiparallel streams & precise phase sync.
   - Result: **Mutual Annihilation** (functioning as a switch) *or* **Fragmentation**. It also induces a space-dependent signal delay.
2. **Glider-Eater Interaction**: These interactions are "less" sensitive to synchronization.
   - Requires: Specific placement/orientation of the Eater.
   - Result: **Glider Annihilation/Absorpbtion** or **fragmentation**.

2.1.3. Reading outputs

Reading the state of a logic gate in the Game of Life is a measurement of **spatial-temporal presence**. The output of a gate is not solely a boolean value, but a **vector**. Since the gliders

only propagate along intercardinal diagonals (NE,NW,SE,SW), the observation window must be placed precisely in the ”firing line” of the logic gate’s exit path.

4×4 **Observation Window.** To detect a ”1” signal (the presence of a glider), the observer must define a static bounding box of 4×4 cells at a specific coordinate (x, y) on the lattice. Because a glider is a dynamic pattern that fits within 4×4 area during its four-generation cycle.

Detection logic. The output is considered to have the assigned `TRUE` value if the correct number of live cells within the 4×4, and the distribution are characteristic of the glider.

Composition. If a gate’s output needs to be routed to a different input, one must apply rotations or reflections to the initial gun configuration to ensure the output vector aligns with the next gate’s input requirements.

Synchorinization. If the observation window is checked at generation T, but the glider arrives at $T + 1$, the `READ` might detect a partial pattern (fragmentation) or an empty box, resulting in a false **0**. Strictly speaking, one could a priori calculate the Manhattan distance the glider travels to ensure the output is sampled only when the signal has had enough generations to reach the observation box.

2.1.4. Initial conditions for the different gates

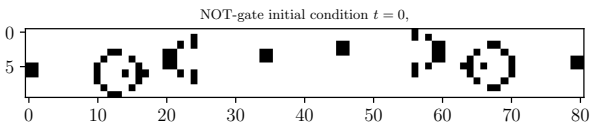

Figure 1. Two Gosper glider guns: the left gun provides a constant input stream, while the input channel injects gliders traveling in the south-east direction. No input is encoded by inserting an eater in the SE input stream. The output is read as gliders propagating in the south-west direction.

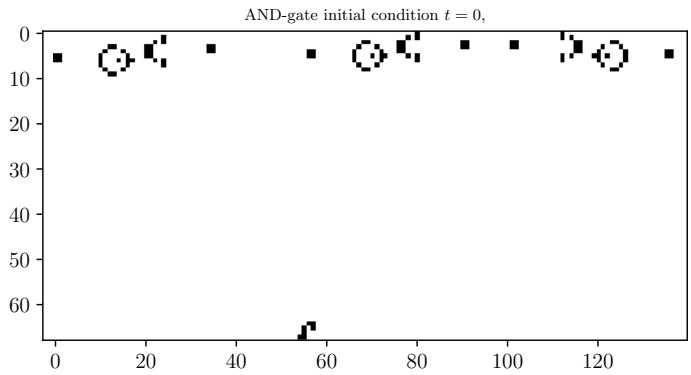

Figure 2. Three Gosper glider guns: the two leftmost generate south-east-directed input streams. An eater suppresses the south-west output unless both inputs are present. In the absence of inputs, additional eaters are placed along the collision paths to block spurious gliders. The output is read as a south-east-directed stream.

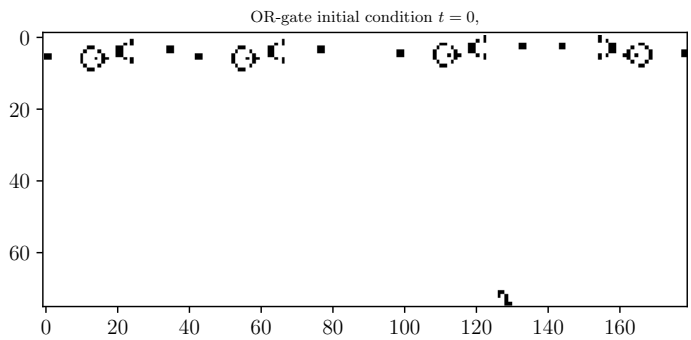

Figure 3. Four Gosper glider guns: the second and third generate south-east-directed input streams. Logical zeros are implemented by placing eaters along the input collision paths. The output is read as a south-east-directed stream.

2.2. Numerical methods for Game of Life

2.2.1. Computational domain

The computational domain is represented as a two-dimensional rectangular lattice $X \in \{0, 1\}^{n \times n}$, where each entry corresponds to one cell of the automaton.

2.2.2. Time evolution

The time evolution is implemented as a composition of a linear convolution operation and a pointwise non-linear update. The discrete convolution is computed using `scipy.ndimage.convolve`. A fixed 3×3 kernel is used to compute the Moore neighbourhood sum $N^{(t)} = K * X^{(t)}$.. The central entry of the kernel is zero, so the cell itself is excluded from the neighbour count. The update function allows both zero-padding boundaries, set by the parameter `mode="constant"`.

The Game of Life update rule is then applied by vectorized logical operations:

$$X^{(t+1)} = \mathbb{1}\left(N^{(t)} = 3\right) \vee \left[X^{(t)} \wedge \mathbb{1}\left(N^{(t)} = 2\right)\right]. \tag{2.5}$$

2.2.3. Initial conditions

Initial conditions are generated programmatically in `GoLogic.py`. The basic building blocks are the Gosper glider gun and the eater, both stored in Run Length Encoding (RLE). A custom parser converts each RLE string into a binary array using `RLE.py` and `add_pattern`. Before adding a pattern one can transform it appropriately with the `transform_pattern` function in `pattern_processing.py`. handle insertion and transformation before simulation.

In addition, the function `load_live_cells(filename, n)` loads external initial configurations from text files containing one live cell per line in the format `(x,y)`. The file is converted into an $n \times n$ binary array by setting `grid[y,x] = 1`. This supports the coordinate-list representation used for the initial gate configurations.

2.2.4. Logical gates

The logical gates `AND`, `OR`, and `NOT` are implemented by spatial arrangements of glider guns and eaters. Signals are encoded as glider streams emitted by Gosper guns. Boolean inputs are represented through the initial configuration: an input value of 1 leaves the corresponding signal unobstructed, whereas an input value of 0 places an eater on the signal path and suppresses propagation.

For the `AND` and `OR` gates, all four input combinations were simulated,

$$(A, B) \in \{(0, 0), (0, 1), (1, 0), (1, 1)\},$$

while the `NOT` gate was tested for both possible input values,

$$A \in \{0, 1\}.$$

The output was read from a designated output lane: the presence of a propagating glider stream was interpreted as logical 1, and its absence as logical 0.

2.3. Game of Life Results

2.4. NOT-gate

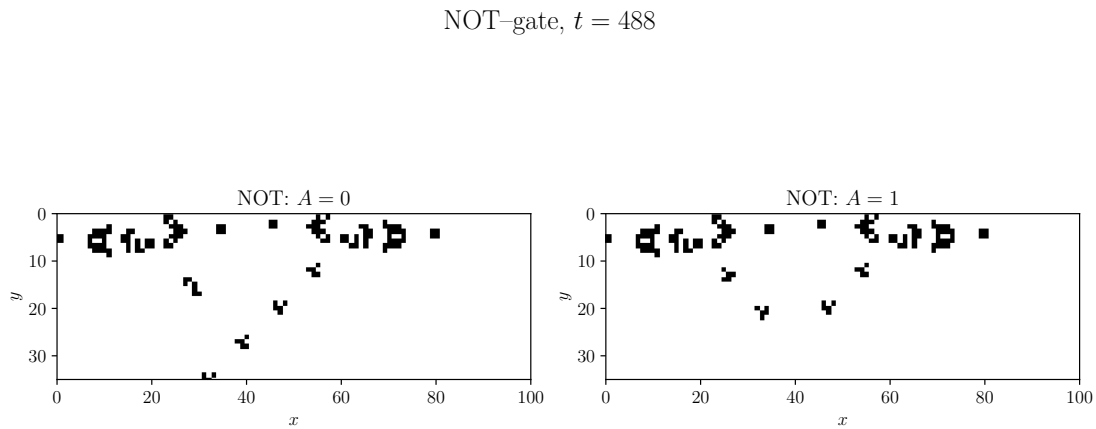

Figure 4. One input channel, A. The input glider moves diagonally from the top to the lower-right part of the grid. The output is observed after the interaction region, in the outgoing diagonal channel. The construction is arranged so that when $A = 0$ an output glider survives, while when $A = 1$ the input glider destroys or blocks the outgoing signal.

2.5. AND-gate

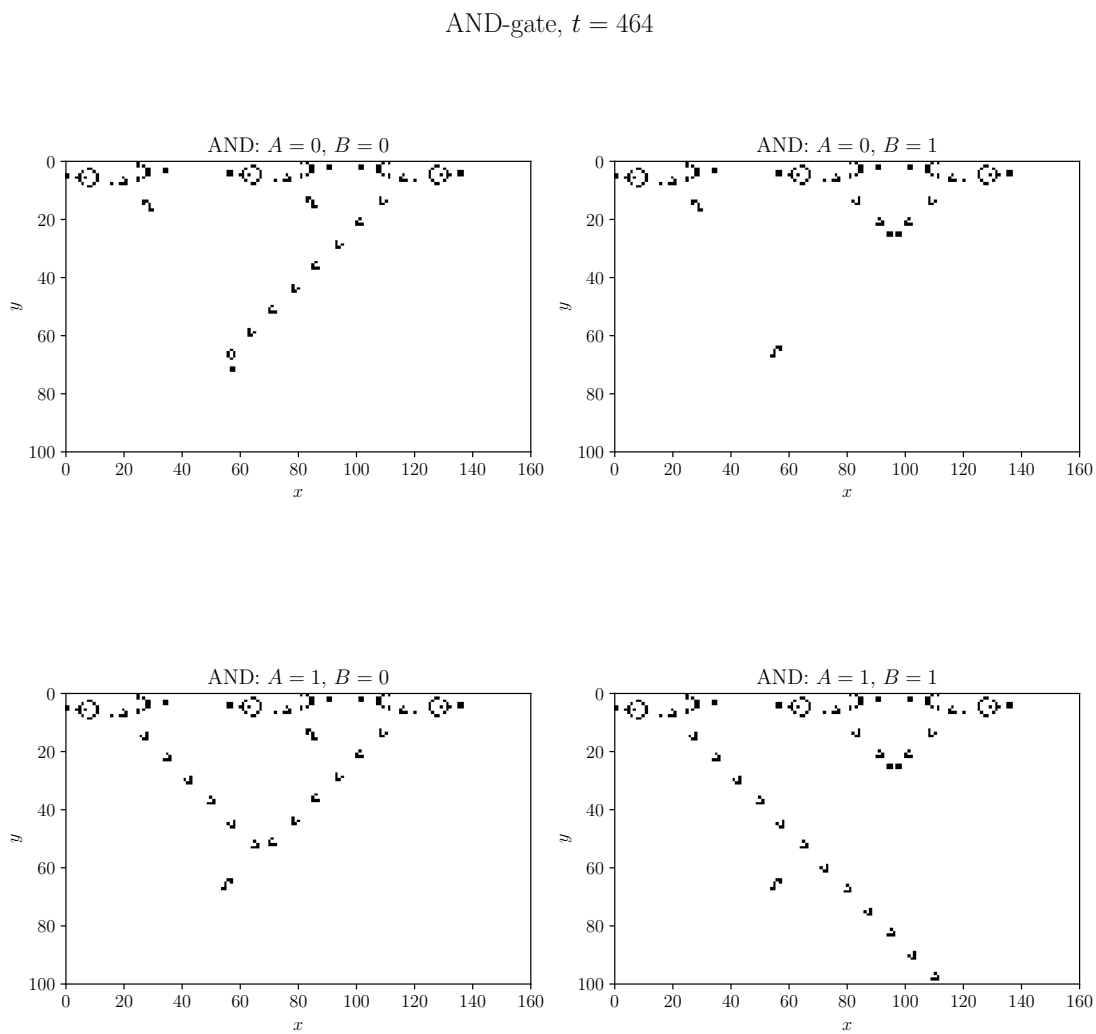

Figure 5. Two input channels are used, A and B. The input gliders move diagonally from the top of the grid towards the lower-right part of the grid, where they interact with the gate structure. The output is observed in the outgoing diagonal channel after this interaction. A glider appears in the output channel only when both input gliders are present, so the output is 1 only for $A = 1$ and $B = 1$.

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2.6. OR-gate

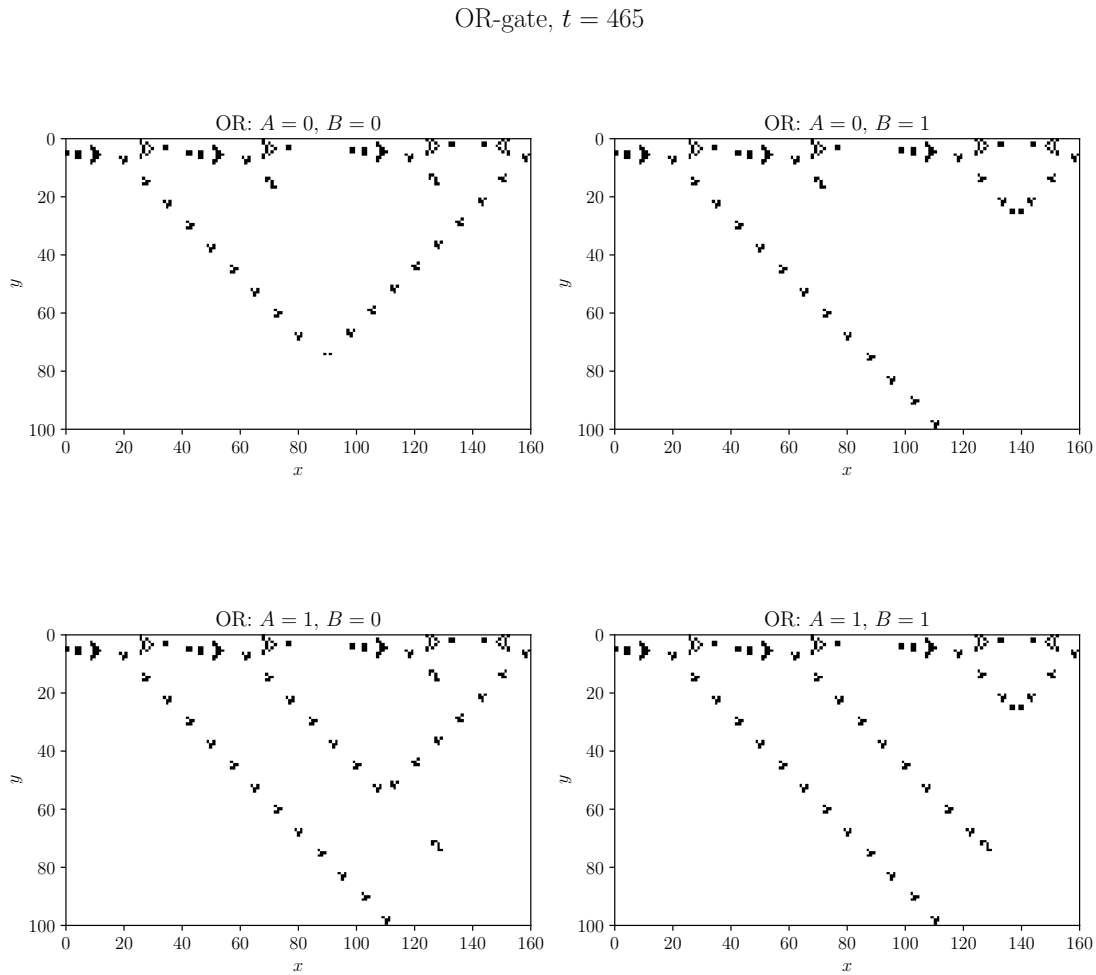

Figure 6. Input gliders also move diagonally from the top of the grid towards the lower-right part of the grid. The output is shown in the outgoing diagonal channel after the region of collision. In this case, the output channel contains a glider when, at least, one of the two input gliders is present.

3. Neuron Cellular Automaton

3.1. Neuron Automaton Theory

FOR THEORY OF GHCA...

3.2. Neuron Cellular Automaton

The script for the implementation of the neuron cellular automaton is `neuron.py`. The code is based on `gameOfLifeInteractiveToric.py`, which was available on the Studium page of the course.

Now, we will present how the code works. At first, the number of rows and columns of the grid as well the speed of the simulation are defined. Additionally, the rules text is added. This text is displayed when clicking the ”rules”-button while running the script. Next, we have implemented the new function `count_neighbors_by_state` that can calculate the number of neighbours in a certain state. This function is necessary as we will be working with three different states instead of two. The function takes the grid and the state that we want to determine the number of neighbours for as arguments. The neighbours for each gridpoint are found using `numpy.roll` to include the periodic boundary conditions. As the next step we have updated the `lifeStep` function. Now, we first create the grids of neighbours in states 1 and 2 and apply the rules afterwards. State 0 represents the resting neurons, state 1 the ready neurons and state 2 the firing neurons. The function then returns the updated grid.

In the constructor of the `GameOfLifeApp` class the visualisation is defined. This includes the buttons that are displayed and their actions. The time step for the continuous run is defined as well.

The following parameters are fixed in the code: the number of rows and columns is set to 30 each. For the random population of the grid, the probability for state 0 is 0.7, for state 1 the probability is 0.2 and 0.1 for state 2. Initially, all cells are in state 0 (resting). Clicking once sets them to state 1 and clicking another time to state 2.

3.3. Neuron Automaton Results

FOR RESULTS OF NEURON AUTOMATON...

4. Greenberg-Hastings Cellular Automaton

4.1. GHCA theory

GHCA THEORY HERE

4.2. GHCA Method

FOR IMPLEMENTATION OF GHCA...

4.3. GHCA Results

FOR RESULTS OF GHCA...

5. Interpretation

INTERPRETATION GOES HERE.

6. Conclusion

In Section (2.3) we presented snapshots illustrating the realization of logic gates within a computer simulation of the Game of Life. From this, one may conclude that a Boolean basis is effectively implemented within the system. Once the principles for composing these gates into larger circuits are understood, increasingly complex computational structures can be constructed.

The existence of logic gates in the Game of Life implies that they form a functionally complete basis for Boolean logic. Consequently, any Boolean function can be expressed as a finite composition of such gates. This property is fundamental for the construction of general computational devices. In particular, it is possible to design patterns that behave as a finite-state machine coupled to two counters. Such a system is computationally equivalent to a universal Turing machine

It follows that the Game of Life possesses the same theoretical computational power as any computer with unbounded memory and no time constraints; that is, it is Turing complete [5]. This demonstrates that complex dynamics can emerge solely from prescribed initial conditions and a simple update rule, forming a discrete dynamical system.

7. Use of LLM tools

7.1. Game of Life

RLE prompt: Create an rle loader to import standard Game of Life patterns directly. The output was a complete functional script `RLE.py` that parses strings like

```
24bo11b$22bobo11b$12b2o6b2...
```

, which encode binary patterns stored in a Wiki page.

7.2. Neuron Cellular Automaton

The code in `neuron.py` is based on `gameOfLifeInteractiveToric.py` from the lecture. We used ChatGPT to modify the code. The prompt was ”I would like to change the rules of the game. I need to have three different states for each cell now and new rules, how can I implement this?” and additionally added the original code. This returned good instructions on where the code needs modification. Additionally, we looked up the documentation of the functions used by ChatGPT and made some modifications to its suggestions.

Files submitted

The following files have been submitted with this report:

1. `GameOfLife.py`
2. `GameOfLifeDebug.py`
3. `GoLogic.py`
4. `pattern_processing.py`
5. `RLE.py`
6. `NOT.txt`
7. `AND.txt`
8. `OR.txt`

References

[1] Alex Bellos. Game of life: Logic gates.

[2] Jordi-Lluís Figueras. Interactive conway's game of life board for classroom demonstrations, 2026.

[3] LifeWiki. Glider.

[4] LifeWiki. Glider gun.

[5] Wikipedia. Turing completeness.